



SIMATS ENGINEERING

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Course Code: CSA1522

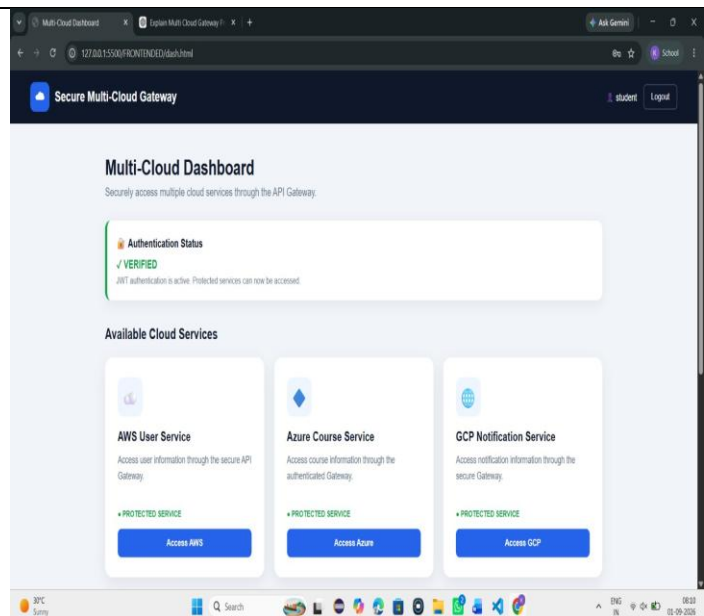
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Course Name: Cloud Computing And Big Data Analytics

Course Faculty: Dr.Rajaram P

Project Title: Web API Gateway For Secure Multi-cloud Application Access

Module Photographs:



Project Description:

Web API Gateway for Secure Multi-Cloud Application Access is a security-focused project designed to provide a centralized and secure access layer for applications and services deployed across multiple cloud platforms. In a multi-cloud environment, organizations often use services from different providers, which can create challenges related to authentication, authorization, API security, traffic management, and monitoring. The proposed system addresses these challenges by introducing an API Gateway that acts as a single entry point between clients and backend applications hosted across different cloud environments. The system authenticates users and applications before allowing access to protected APIs. It can implement **token-based authentication using JWT**, role-based access control (RBAC), API key validation, and secure communication through HTTPS. Based on the authenticated user's role and access permissions, the gateway forwards requests only to authorized services. This helps prevent unauthorized access and reduces the direct exposure of cloud-hosted backend services.

Student Signature

Guide Signature